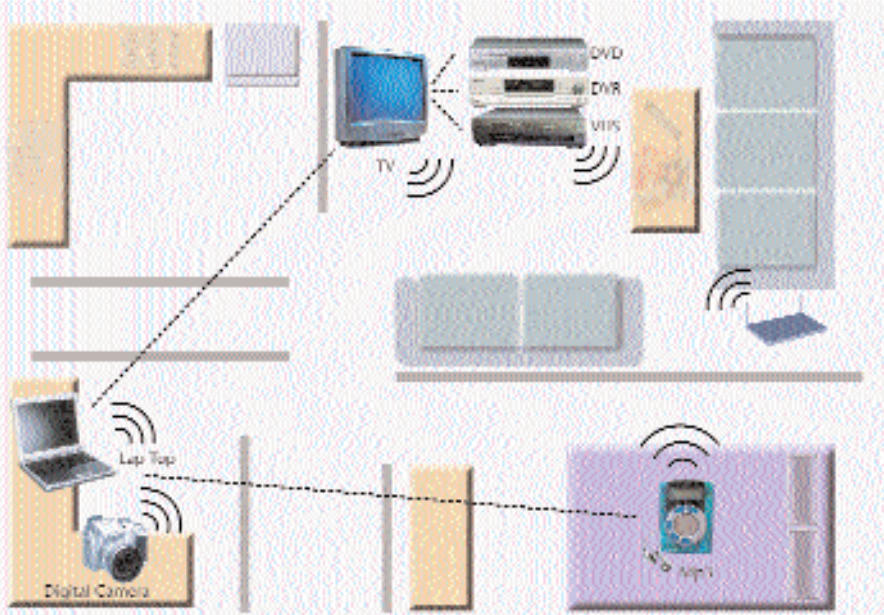


A "mesh" networking topology within the home



Ultra Wideband, as the wireless transmission technology used in this architecture, allows for very low-cost and low-power access points within the home. Existing wireless solutions simply cannot meet the cost and data rate requirements of such an access point

Bruce W. Watkins, Pulse~LINK President and Chief Operating Officer, corroborates: "There is no FCC limiting factor of 10 meters—this is a very common misconception. Nothing in the FCC ruling or the world of physics limits the technology to 10 metres." It is the wireless UWB signal modulation techniques used that limits a company's particular implementations of UWB to 10 metres, he adds.

How UWB works

UWB technology has its origins in espionage agencies in both the US and the former Soviet Union. In the late 1970s, the research of military radar technicians led to the 'ultra wideband synthetic aperture radar', used by spy planes and satellites to see through dense ground cover to locate enemy troops and camouflaged equipment on the ground. It works by showering the target with rapid pulses of broad, low-frequency signals that punch their way through solid objects.

UWB works in a similar fashion. Rather than broadcasting on separate government-auctioned frequencies, UWB blasts signals across the entire radio spectrum. The typical long radio wave is replaced by rapid fire Morse code-like bursts, sending hundreds of low-power (around 50 to 70 milliwatts of power—

ten-thousandth the power of a cellphone) electrical pulses each second, each lasting under a nanosecond.

By varying the pulse timing (see box, 'Catching the UWB Pulse') according to a complex code, a pulse can represent either a zero or a one: the basis of digital communications. Because the signal is so low power and so broadly spread, it is both subject to, and subjects other devices to, little or no interference. Only a UWB receiver that knows the exact pulse sequence generated by the transmitted signal can make use of the information in the signal. To the rest of the world it is just more background noise.

UWB energy pulses operate in the same frequency spectrum as electronic 'noise' emitted by Pentium chips, TV monitors, electric razors, automobile ignitions and fans. This is a huge swath of the spectrum, regulated only by FCC rules on how much power these devices can use.

Low power and the characteristic wide spread of the pulses means the pulses don't use up already crowded chunks of the radio spectrum, today occupied by 802.11b wireless LANs and Bluetooth devices. Hence, UWB would allow a whole new class and volume of voice and data communications that, in effect, wouldn't take up any more 'space' in the crowded radio spectrum.

UWB transmitters and receivers are also said to be simpler to build, run and maintain than those in use today. For UWB, you don't need complex radio frequency converters and modulators. You only need a digital method to construct the pulses and modulate them. This can all go on a single chip.

How secure is UWB?

UWB promises to be highly secure. It's very difficult to filter a pulse signal out of the flood of background electronic noise with traditional RF scanners. Besides, vendors such as Time Domain are encrypting the zeros and ones being transmitted by the pulses.

Watkins maintains that even if an 'intruder' could find the signal in the noise floor, maintaining precise synchronisation with a series of pseudo-random pulses of energy present less than 10 per cent of the time and less than 500 picoseconds in duration is a monumental engineering challenge.

Applications

Now that the FCC has given its blessing to the use of UWB technology for wireless networking, companies such as Time Domain and XtremeSpectrum are trying to bring consumer electronics manufacturers on board. Time Domain is already working with more than 24 companies to develop applications for its UWB chipsets.

Leading consumer-electronics firms are expected to begin embedding UWB chipsets into DVD players, VCRs, video-game consoles and home-theatre equipment in the coming years. The home-networking market is expected to reach \$4 billion by 2006.

Last July, XtremeSpectrum demonstrated the extreme bandwidth and 'wire-like' video quality of its new Trinity chipset, a wireless semiconductor solution suited for applications in the consumer connectivity market. Using the popular MPEG2 video format, the company broadcast six video streams to six separate flat panel displays simultaneously across the room using a single UWB connection.

Philips and General Atomic teamed up to produce UWB chipsets that could help the home network run 40 times faster than today's IEEE 802.11b networks. Philips envisions the chipsets that transfer digital content at 480 Mbps between clusters of entertainment or computing devices in the home. These could include digital video recorders, set-top boxes, TVs, and PCs. Last